8 - Linear Regression IV

Exogeneity and Omitted Variables

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Agenda

Today we ask: When can a regression coefficient represent the population relationship or effect we care about?

We will cover:

1. The error term and zero conditional mean
2. Exogeneity and what it means substantively
3. Omitted-variable bias and confounding
4. Why controls help, but do not guarantee causality

Recommended reading: Wooldridge, Chapters 3.1–3.3

1. The Credibility Problem

What is Still Hiding in the Error Term?

Suppose we estimate:

$$Price = \beta_0 + \beta_1 RAM + \beta_2 CPUfreq + \beta_3 Apple + u$$

We can interpret $\hat{\beta}_1$ as the association between *RAM* and *Price*, holding *CPUfreq* and *Apple* fixed.

But a causal interpretation requires that, once we hold the included controls fixed, RAM does not systematically predict the average value of the error term.

So the key question becomes:

What is still hiding in u?

What Is the Error Term Doing?

Recall that u contains all determinants of y that are **not explicitly included** in the regression model.

For a laptop-pricing model, u may contain:

- build quality;
- product segment;
- screen quality;
- launch year;
- temporary discounts;
- unobserved demand conditions.

The problem is not that u exists.

The **problem for causal interpretation** is when the explanatory variables systematically predict whether u tends to be positive or negative on average.

MLR.4: Zero Conditional Mean

The **key assumption** is:

$$E[u \mid x_1, x_2, \dots, x_k] = 0$$

In words:

> *Once we condition on the included explanatory variables, the average value of the unobserved factors is zero.*

Equivalently:

> *Knowing the values of the included explanatory variables should not help us predict whether the error term is positive or negative* ***on average****.*
>
> *This is a condition about the average of u; it does not require x and u to be completely independent.*

Exogeneity: The Fair-Comparison Assumption

An explanatory variable is **exogenous** when, conditional on the other included variables, it does not systematically predict the average value of u.

For x_j, the concern is:

$$x_j \quad \text{predicts the average of} \quad u?$$

- If yes, observations with different values of x_j also systematically differ in unobserved determinants of y.
- That is why the estimated coefficient may mix together multiple effects.

2. Confounding in Action

Case: Do Storks Deliver Babies?

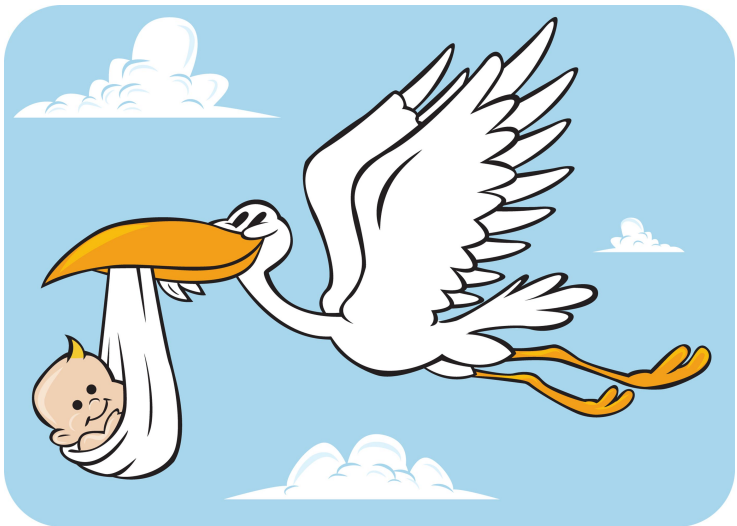
Storks have long been associated with babies.

Suppose we collect country-level data and ask:

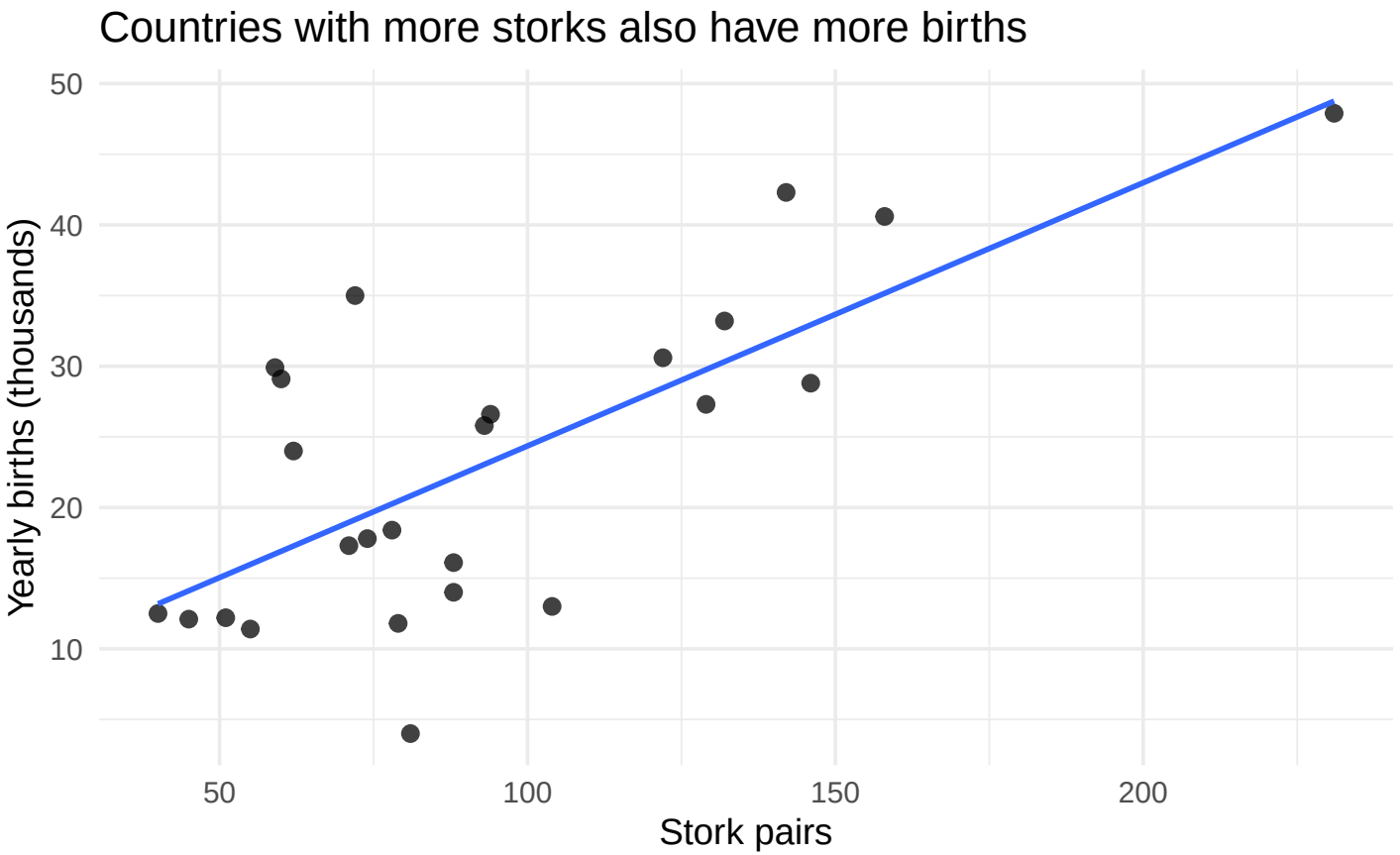
> ***Are countries with more stork pairs also countries with more births?***

Before looking at the regression:

- What is the **outcome**?
- What is the **explanatory variable**?
- What could be a **confounder**?
- Would a positive coefficient be **causal evidence**?



Storks and Births: The Naive Relationship



Naive Regression Result

```
# Naive regression
stork_slr <- lm(yearly_birth_rate_thousands ~ stork_pairs, data = storks)
```

Variable	Estimate	SE	p-value
stork pairs	0.186	0.037	$< .001$

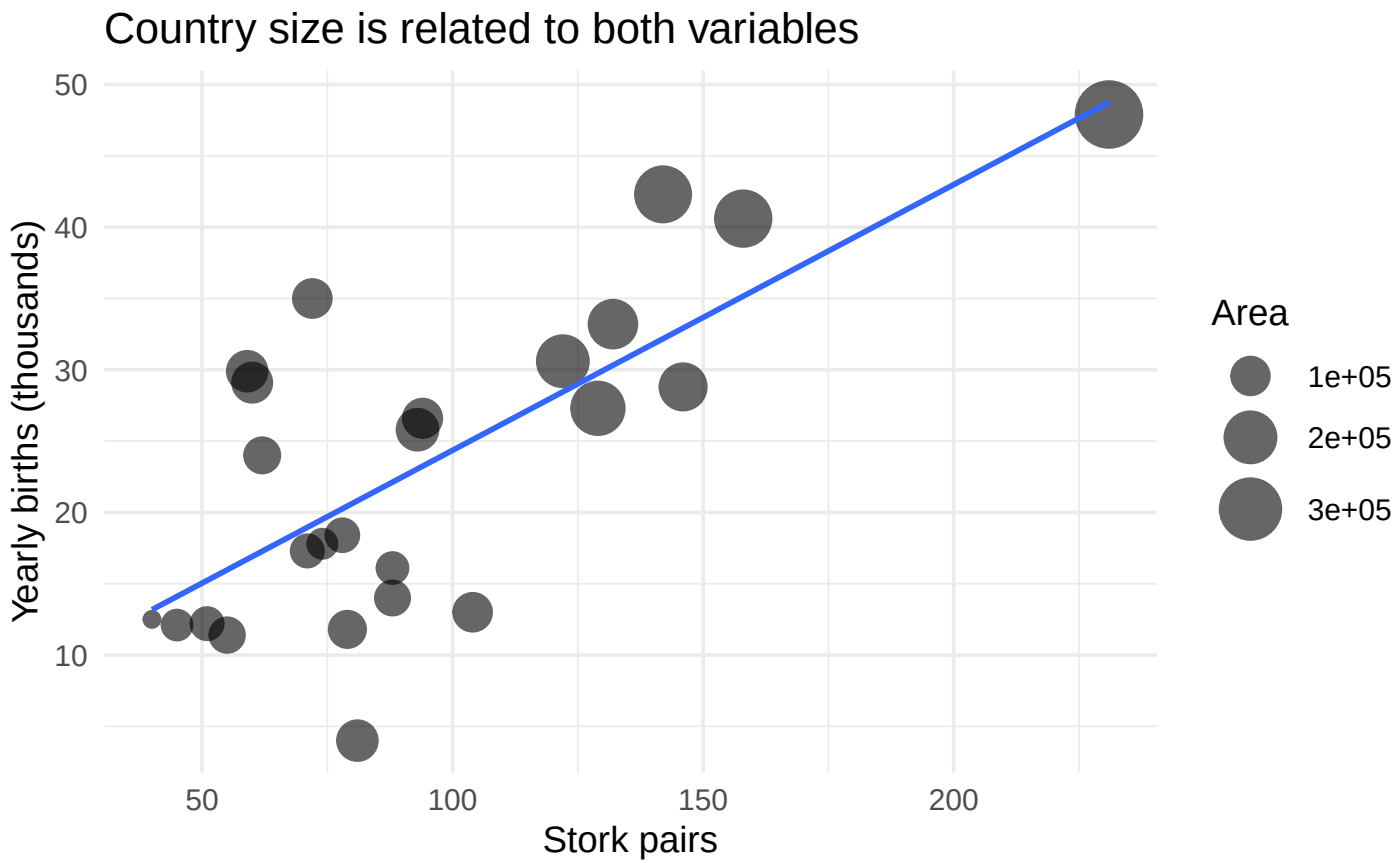
We could conclude that:

> *Countries with one additional stork pair have, on average, about 0.186 thousand more births per year — approximately 186 additional births per year.*

But this is **not** a credible causal interpretation.

What Is Hiding in u?

One obvious omitted factor is **country size**.



Controlling for the Confounder

Now estimate:

$$births = \beta_0 + \beta_1 storks + \beta_2 area + u$$

```
# Regression with control for area
stork_area <- lm(yearly_birth_rate_thousands ~ stork_pairs + area_km2, data = storks)
```

Variable	Estimate	SE	p-value
storks only	0.186	0.037	$< .001$
storks controlling for area	-0.048	0.087	0.586

Once we control for country size, the stork coefficient no longer carries the same interpretation - both magnitude and significance change.

The Confounding Story

The problem can be summarized as:

$$\text{area} \longrightarrow \text{storks}$$

$$\text{area} \longrightarrow \text{births}$$

If area is omitted, it becomes part of the error term:

$$u = \text{area effect} + \text{other unobserved factors}$$

Because area affects births and is related to storks, the part of u generated by omitted area varies systematically with the number of storks.

$$E(u \mid storks) \neq 0$$

3. Omitted-Variable Bias

When Does Omitted-Variable Bias Arise?

Omitted-variable bias requires two ingredients.

A variable is problematic if it is:

1. a determinant of the outcome y;
2. correlated with the included explanatory variable of interest.

If the true model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$$

but we estimate:

$$y = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + v$$

then x_2 is omitted and becomes part of the new error term v.

Direction of Omitted-Variable Bias

In the simple omitted-variable case:

$$Bias(\tilde{\beta}_1) = \beta_2 \delta_1$$

where δ_1 is the slope from:

$$x_2 = \delta_0 + \delta_1 x_1 + r$$

So the sign of the bias depends on:

- the effect of the omitted variable on y;
- the relationship between the omitted variable and x_1.

Bias Sign Table

	$\text{Corr}(x_1, x_2) > 0$	$\text{Corr}(x_1, x_2) < 0$
x_2 raises y ($\beta_2 > 0$)	Positive bias	Negative bias
x_2 lowers y ($\beta_2 < 0$)	Negative bias	Positive bias

Application to storks:

- area raises expected births;
- area is positively related to stork pairs;
- omitting area creates **positive bias** in the stork coefficient.

If an explanatory variable is correlated with the error term, MLR.4 is violated; in multiple regression, this can generally bias more than just that variable's coefficient.

4. What Controls Can and Cannot Do

Controls Help When They Improve the Comparison

A useful control variable should help compare observations that are more similar in ways relevant to the outcome.

Good controls are usually variables that:

- affect y;
- are related to the explanatory variable of interest;
- are measured before or independently of the outcome;
- are conceptually justified by the research question.

Controls should come from theory, institutional knowledge, business understanding, and careful thinking about the data-generating process.

Controls Do Not Guarantee Causality

Even after controls, causal interpretation still requires:

$$E(u \mid x_1, x_2, \ldots, x_k) = 0$$

Substantively, this requires us to think carefully about whether:

- no important omitted confounders remain;
- included controls are measured appropriately;
- the model captures the relevant comparison;
- the explanatory variable is not jointly determined with y.

These are substantive assumptions, not things software can verify for us.

What Can We Check? What Can We Argue?

Software can help with:

- model output, residual patterns, fit, sensitivity to added controls

Judgment is needed for:

- whether the comparison is fair, which variables are confounders, whether causal language is justified

We cannot prove exogeneity from the data alone.

But we can make a more or less convincing argument based on the question, the data-generating process, and robustness checks.

A Practical Checklist Before Causal Language

Before saying "effect" or "impact", ask:

1. What is the comparison behind the coefficient?
2. What important factors are still in u?
3. Do those factors tend to vary systematically with the explanatory variable?
4. Have we included the most relevant and appropriate controls?
5. Is there a credible reason to believe that $E(u \mid X) = 0$ - that the included explanatory variables do not systematically predict the average error?

If not, use associational language.

5. Closing

Key Takeaways

- Multiple regression compares observations **holding included variables constant**.
- Causal interpretation requires that, conditional on the included variables, the explanatory variables do not systematically predict the average error.

$$E(u \mid X) = 0$$

- Omitted-variable bias arises when an omitted factor affects y and is related to an included explanatory variable.
- Controls can improve the comparison, but they do not automatically solve the credibility problem.

The central question is always: what is hiding in u?

Where This Leaves Us

We now have the main logic of regression:

$$\text{question} \rightarrow \text{model} \rightarrow \text{coefficient} \rightarrow \text{uncertainty} \rightarrow \text{credibility of assumptions}$$

The coefficient is only the beginning.

The harder task is deciding what the coefficient can reasonably mean.

Next: Revision and Q&A before the midterm.

Recommended Readings

Wooldridge, *Introductory Econometrics: A Modern Approach*, 7th ed.

- Chapter 3.1 Motivation for Multiple Regression
- Chapter 3.2 Mechanics and Interpretation of Ordinary Least Squares
- Chapter 3.3 The Expected Value of the OLS Estimators
  - Section 3.3b Omitted Variable Bias: The Simple Case
  - Section 3.3c Omitted Variable Bias: More General Cases